

CMOS VLSI Design Project

Design and Analysis of a 2-Input CMOS NOR Gate

Using Cadence Virtuoso

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1 Introduction

The CMOS NOR gate is one of the fundamental building blocks of digital integrated circuits. It performs the logical NOR operation, where the output becomes HIGH only when all input signals are LOW.

In this project, a two-input CMOS NOR gate was designed and analyzed using the Cadence Virtuoso design environment. The complete custom CMOS design flow included schematic design, symbol generation, physical layout design, Design Rule Check (DRC), Layout Versus Schematic (LVS) verification, RC parasitic extraction (RCX), and post-layout simulation.

The post-layout simulation was used to evaluate important performance parameters such as average power consumption and propagation delay.

The Boolean expression of a two-input NOR gate is

$$Y = \overline{A + B} \quad (1)$$

where A and B represent the input signals and Y represents the output.

2 Objectives

The main objectives of this project are:

- To design a two-input CMOS NOR gate using Cadence Virtuoso.
- To create and verify the transistor-level schematic of the NOR gate.
- To generate a reusable symbol for the designed NOR gate.
- To design the physical layout of the CMOS NOR gate.
- To verify the physical layout using Design Rule Check (DRC).
- To verify the electrical equivalence between the schematic and layout using Layout Versus Schematic (LVS).
- To perform parasitic resistance and capacitance extraction using RCX.
- To perform post-layout transient simulation using the extracted circuit.
- To calculate the average power consumption of the NOR gate.
- To calculate the propagation delay of the NOR gate.

3 Theory of CMOS NOR Gate

A two-input CMOS NOR gate consists of two PMOS transistors forming the pull-up network and two NMOS transistors forming the pull-down network.

The two PMOS transistors are connected in series between V_{DD} and the output node. The two NMOS transistors are connected in parallel between the output node and ground.

The input signals A and B control the gates of the corresponding PMOS and NMOS transistors.

The Boolean expression of the NOR gate is

$$Y = \overline{A + B}. \quad (2)$$

The truth table of the two-input CMOS NOR gate is shown in Table 1.

Table 1: Truth Table of a Two-Input CMOS NOR Gate

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

The output becomes HIGH only when both input signals are LOW.

When $A = 0$ and $B = 0$, both PMOS transistors turn ON, producing a conducting path from V_{DD} to the output node. At the same time, both NMOS transistors turn OFF. Therefore, the output becomes HIGH.

When either A or B becomes HIGH, the corresponding PMOS transistor turns OFF and the corresponding NMOS transistor turns ON. Since the NMOS devices are connected in parallel, this creates a conducting path between the output node and ground. Therefore, the output becomes LOW.

Hence, the implemented circuit performs the required NOR operation:

$$Y = \overline{A + B}. \quad (3)$$

4 Transistor Sizing

The CMOS NOR gate was designed using two PMOS transistors and two NMOS transistors.

From the designed schematic, the approximate transistor dimensions used in the circuit are:

Table 2: Transistor Dimensions Used in the CMOS NOR Gate

Transistor	Number	Width	Length
PMOS	2	480 nm	100 nm
NMOS	2	240 nm	100 nm

Therefore, the PMOS-to-NMOS width ratio used in the design is

$$\frac{W_p}{W_n} = \frac{480}{240} = 2. \quad (4)$$

The PMOS transistors were designed with approximately twice the width of the NMOS transistors. This larger PMOS width helps provide adequate pull-up strength and compensates for the lower mobility of holes relative to electrons.

5 Circuit Design

5.1 Schematic Design

The two-input CMOS NOR gate schematic was created using the Cadence Virtuoso Schematic Editor.

The circuit consists of two PMOS transistors connected in series between V_{DD} and the output node, while two NMOS transistors are connected in parallel between the output node and ground.

The input terminals are represented by A and B , while the final output terminal is represented by Y .

The schematic of the designed NOR gate is shown in Figure 1.

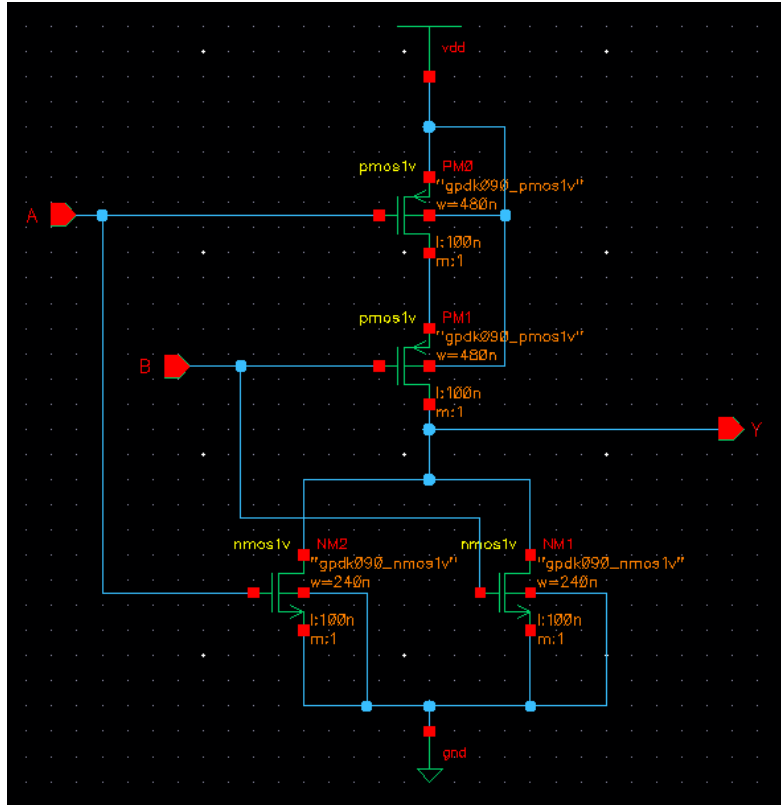


Figure 1: Schematic of the two-input CMOS NOR gate in Cadence Virtuoso.

The logical operation of the circuit can be represented as

$$A, B \rightarrow \overline{A + B}. \quad (5)$$

5.2 NOR Gate Symbol

After completing the schematic, a symbol was generated for the two-input CMOS NOR gate.

The symbol contains two input terminals labeled A and B and one output terminal labeled Y .

The generated NOR gate symbol is shown in Figure 2.

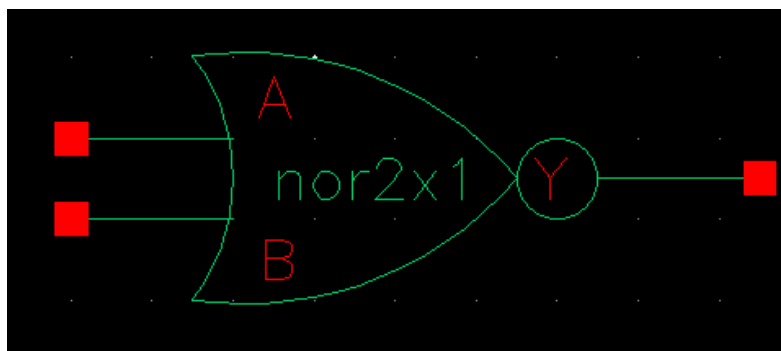


Figure 2: Generated symbol of the two-input CMOS NOR gate.

The symbol allows the designed NOR gate to be used conveniently as a reusable hierarchical block in larger digital circuits.

6 Physical Layout Design

After completing the schematic and symbol generation, the physical layout of the CMOS NOR gate was developed using Cadence Virtuoso Layout Editor.

The PMOS transistors were placed in the upper portion of the layout, while the NMOS transistors were placed in the lower portion.

The physical layout contains the required layers for transistor formation and electrical interconnection, including:

- active or diffusion regions,
- polysilicon,
- N-well,
- metal layers,
- contacts,
- vias,
- power connection, and
- ground connection.

The PMOS devices form the series pull-up network, while the NMOS devices form the parallel pull-down network.

The input signals are routed to the appropriate transistor gates, and the output node is connected between the pull-up and pull-down networks.

The completed physical layout is shown in Figure [3](#).

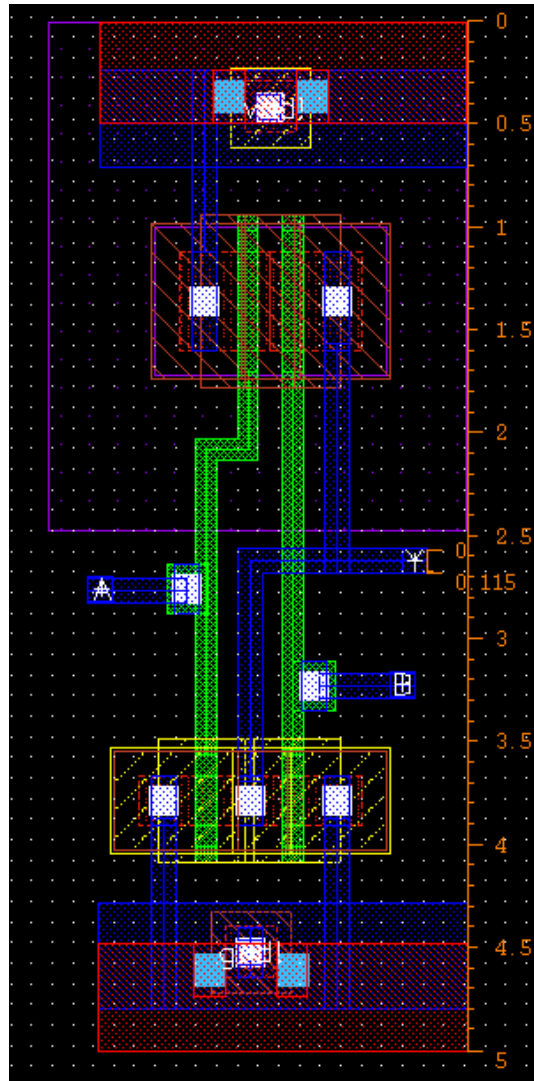


Figure 3: Physical layout of the two-input CMOS NOR gate.

The layout was designed while maintaining the required minimum width, spacing, enclosure, and connectivity rules of the selected CMOS technology.

7 Design Rule Check (DRC)

Design Rule Check (DRC) is performed to verify whether the physical layout satisfies the fabrication rules of the selected CMOS technology.

DRC checks various geometrical constraints such as:

- minimum metal width,
- minimum metal spacing,
- minimum polysilicon width,
- polysilicon spacing,

- diffusion spacing,
- contact dimensions,
- contact enclosure,
- via enclosure, and
- well spacing.

The completed NOR gate layout was checked using the Cadence DRC verification flow.

DRC Result: PASSED

The physical layout satisfies the required design rules of the selected CMOS technology.

Thus, the physical layout is valid with respect to the required fabrication design rules.

8 Layout Versus Schematic (LVS)

After successful DRC verification, Layout Versus Schematic (LVS) verification was performed.

LVS compares the electrical netlist extracted from the physical layout with the netlist generated from the original schematic.

It verifies important circuit properties such as:

- number of MOS transistors,
- PMOS and NMOS device types,
- gate connectivity,
- source and drain connectivity,
- input connections,
- output connection,
- power connection, and
- ground connection.

The final NOR gate layout was checked against the original schematic.

LVS Result: PASSED

The physical layout is electrically equivalent to the original schematic.

The successful LVS result confirms that the intended electrical connectivity was maintained during the physical layout design.

9 RCX and Parasitic Extraction

After DRC and LVS verification, RC parasitic extraction was performed using the RCX extraction flow.

Physical interconnections in an integrated circuit introduce additional parasitic resistance and capacitance.

These parasitic elements can affect important performance parameters such as propagation delay and power consumption.

RCX extracts these parasitic resistance and capacitance components from the physical layout.

The extracted view of the designed NOR gate is shown in Figure 4.

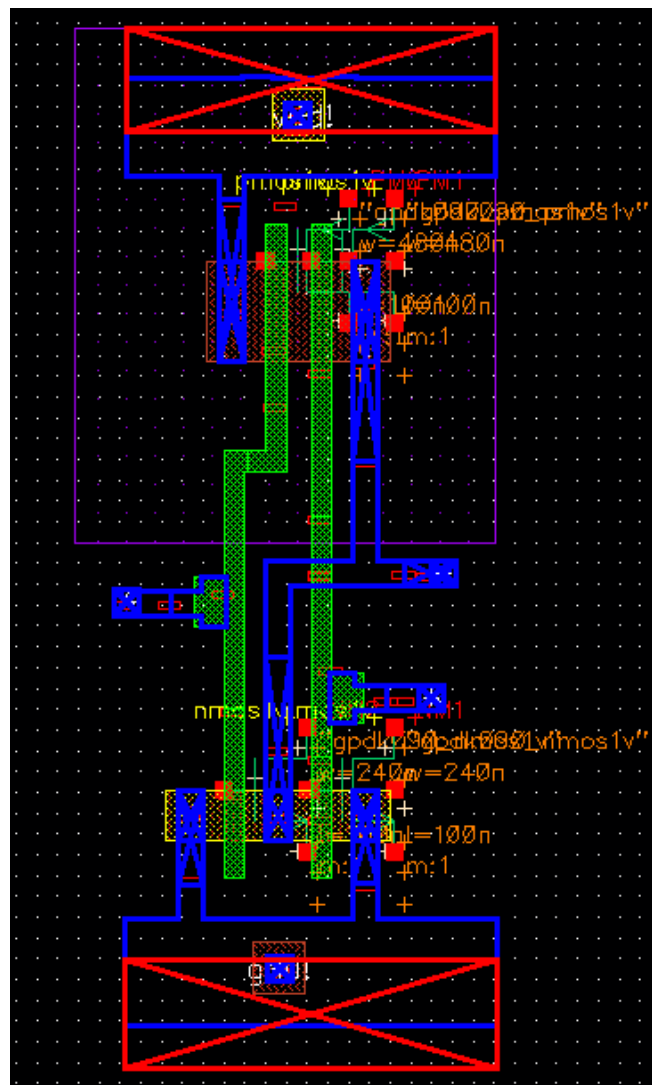


Figure 4: RCX/extracted view of the two-input CMOS NOR gate.

RCX: Successfully Completed

The extracted NOR gate circuit was used for post-layout simulation.

10 Post-Layout Simulation

After parasitic extraction, post-layout transient simulation was performed using the RC-extracted view of the NOR gate.

Unlike ideal schematic-level simulation, post-layout simulation includes the effects of parasitic resistance and capacitance generated by the physical layout.

Therefore, post-layout simulation provides a more realistic estimation of the circuit performance.

The main performance parameters calculated in this project were:

1. Average power consumption
2. Propagation delay

11 Average Power Calculation

The average power consumption of the NOR gate was determined from the post-layout transient simulation results using the Cadence Virtuoso Calculator.

The average power over a time interval T can be expressed as

$$P_{\text{avg}} = \frac{1}{T} \int_0^T p(t) dt \quad (6)$$

where $p(t)$ represents the instantaneous power consumption.

The average value of the simulated power waveform was calculated using the Cadence Virtuoso Calculator.

The power calculation is shown in Figure 5.

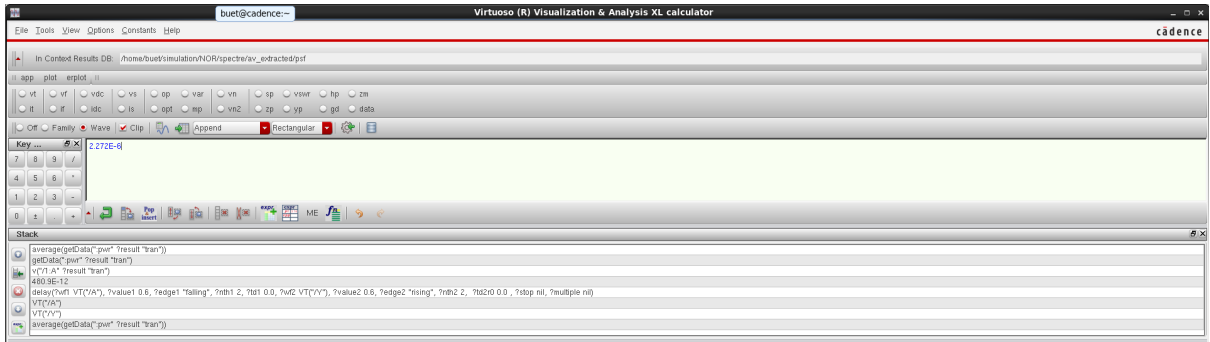


Figure 5: Average power calculation of the extracted CMOS NOR gate using the Cadence Virtuoso Calculator.

From the Cadence Calculator, the obtained average power is

$$P_{\text{avg}} = 2.272 \times 10^{-6} \text{ W.} \quad (7)$$

Since

$$1 \mu\text{W} = 10^{-6} \text{ W}, \quad (8)$$

the average power consumption can be written as

$$P_{\text{avg}} = 2.272 \mu\text{W} \quad (9)$$

Therefore, the post-layout average power consumption of the designed two-input CMOS NOR gate is approximately

$$2.272 \mu\text{W}. \quad (10)$$

12 Propagation Delay Calculation

Propagation delay represents the amount of time required for an input transition to produce the corresponding transition at the output of a digital circuit.

For the NOR gate, an input transition that causes the output to change is used to determine the propagation delay.

The propagation delay can generally be expressed as

$$t_p = t_{\text{out}} - t_{\text{in}}. \quad (11)$$

In this simulation, a switching reference level of approximately 0.6 V was used.

The Cadence Virtuoso Calculator was used to determine the delay between the selected input transition and the corresponding output transition.

The delay calculation is shown in Figure 6.



Figure 6: Propagation-delay calculation of the extracted CMOS NOR gate using the Cadence Virtuoso Calculator.

The measured propagation delay obtained from the Cadence Calculator is

$$t_p = 480.9 \times 10^{-12} \text{ s}. \quad (12)$$

Since

$$1 \text{ ps} = 10^{-12} \text{ s}, \quad (13)$$

the propagation delay becomes

$$t_p = 480.9 \text{ ps} \quad (14)$$

Therefore, the measured post-layout propagation delay for the selected input-output transition of the CMOS NOR gate is approximately

$$480.9 \text{ ps}. \quad (15)$$

13 Verification and Performance Summary

The overall design, verification, and performance results of the two-input CMOS NOR gate are summarized in Table 3.

Table 3: Summary of the CMOS NOR Gate Design Results

Parameter	Result
Logic Function	$Y = \overline{A + B}$
CMOS Implementation	Two PMOS in series and two NMOS in parallel
PMOS Width	480 nm
NMOS Width	240 nm
Channel Length	100 nm
Schematic Design	Completed
Symbol Generation	Completed
Physical Layout	Completed
DRC	Passed
LVS	Passed
RCX / Parasitic Extraction	Completed successfully
Post-layout Simulation	Completed
Average Power	2.272 μW
Propagation Delay	480.9 ps

14 Discussion

The complete custom CMOS design flow for a two-input NOR gate was carried out using Cadence Virtuoso.

The transistor-level circuit consists of two PMOS transistors forming the pull-up network and two NMOS transistors forming the pull-down network.

The PMOS transistors are connected in series, while the NMOS transistors are connected in parallel. This topology ensures that the output becomes HIGH only when both inputs are LOW.

When either input becomes HIGH, the corresponding NMOS transistor turns ON and creates a conducting path from the output node to ground. Therefore, the output becomes LOW.

The transistor dimensions used in the design were approximately 480 nm for each PMOS transistor and 240 nm for each NMOS transistor, with a channel length of 100 nm.

After completing the schematic, a reusable NOR gate symbol was generated. The physical layout was then developed using the required active, polysilicon, well, metal, contact, via, power, and ground layers.

The layout was checked using Design Rule Check and Layout Versus Schematic verification.

RC parasitic extraction was then performed to obtain the additional resistance and capacitance associated with the physical layout. The extracted circuit was subsequently used for post-layout transient simulation.

The average post-layout power consumption obtained from the Cadence Virtuoso Calculator was approximately

$$P_{\text{avg}} = 2.272 \mu\text{W}. \quad (16)$$

The measured propagation delay for the selected input-output transition was approximately

$$t_p = 480.9 \text{ ps}. \quad (17)$$

These results include the effects of layout-dependent parasitic elements and therefore provide a more realistic indication of the circuit performance than an ideal schematic-level simulation.

15 Conclusion

A two-input CMOS NOR gate was successfully designed, implemented, and analyzed using Cadence Virtuoso.

The required Boolean function was

$$Y = \overline{A + B}. \quad (18)$$

The NOR gate was implemented using two PMOS transistors in series and two NMOS transistors in parallel.

The project covered the major stages of the custom CMOS VLSI design flow, including schematic design, symbol generation, physical layout design, DRC verification, LVS verification, RC parasitic extraction, and post-layout simulation.

The physical layout was checked against the required design rules and its electrical connectivity was verified against the original schematic.

RCX was completed successfully, allowing the effects of parasitic resistance and capacitance to be included in the final simulation.

The obtained post-layout average power consumption was

$$\boxed{P_{\text{avg}} = 2.272 \mu\text{W}} \quad (19)$$

and the measured propagation delay was

$$\boxed{t_p = 480.9 \text{ ps.}} \quad (20)$$

Therefore, the project demonstrates the complete schematic-to-layout design, verification, extraction, and performance analysis of a two-input CMOS NOR gate using Cadence Virtuoso.

16 Software Used

The following software tools were used to complete the project:

- Cadence Virtuoso Schematic Editor
- Cadence Virtuoso Layout Editor
- Cadence Spectre Simulator
- Design Rule Check (DRC) Tool
- Layout Versus Schematic (LVS) Tool
- RCX / Parasitic Extraction Tool
- Cadence Virtuoso Visualization and Analysis Calculator